Integer-Valued Polynomials on Noncommutative Algebras Need Not Form a Ring

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Abstract

We resolve an open problem [2] in commutative ring theory and prove that integer-valued polynomials on noncommutative algebras need not form a ring. The proof is generated by GPT 5.5 Pro and verified by Claude Opus 4.8 ultracode.

1 Introduction

Let D be an integral domain with quotient field K. Following Problem 27 of Cahen–Fontana–Frisch–Glaz [1] (contributed by N. J. Werner), let A be a torsion-free D-algebra that contains D, is finitely generated as a D-module, and satisfies $K \cap A = D$; set $B = K \otimes_D A$. With a *central* indeterminate X and *right evaluation* $f(r) = \sum_i c_i r^i$ (coefficients on the left), one considers

$$\mathrm{Int}(A) = \{\, f \in B[X] \;:\; f(A) \subseteq A \,\}.$$

The set $\mathrm{Int}(A)$ is always a left A-module, but when A is noncommutative it is not clear that it is closed under multiplication, hence not clear that it is a ring (see Werner [3] for the noncommutative integer-valued-polynomial framework). There are two questions in Problem 27 of [2]:

- **(a)** Give an example of a D-algebra A with $\mathrm{Int}(A)$ not a ring (possibly relaxing finite generation); and
- **(b)** Werner [4] conjectured that $\mathrm{Int}(A)$ is always a ring when D has finite residue rings, prove the conjecture or give a counterexample.

We resolve both, with a single construction.

2 Main Proof

Before stating our proof, we establish the notations first. Let $S = \mathbb{F}_2[\varepsilon]/(\varepsilon^4)$. Inside $M_2(S)$, define

$$R = \left\{ \begin{pmatrix} \alpha + \eta\varepsilon^2 & \gamma\varepsilon + \lambda\varepsilon^3 \\ \delta\varepsilon + \mu\varepsilon^3 & \beta + \theta\varepsilon^2 \end{pmatrix} : \alpha, \beta, \gamma, \delta, \eta, \theta, \lambda, \mu \in \mathbb{F}_2 \right\}. \tag{2.1}$$

We use the following elements of R:

$$e = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

$$u = \begin{pmatrix} 0 & \varepsilon \\ 0 & 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 0 & 0 \\ \varepsilon & 0 \end{pmatrix}.$$

Define

$$p = uv, \qquad q = vu, \qquad s = uvu, \qquad w = vuv.$$

Explicitly,

$$p = \begin{pmatrix} \varepsilon^2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad q = \begin{pmatrix} 0 & 0 \\ 0 & \varepsilon^2 \end{pmatrix},$$

and

$$s = \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 0 \\ \varepsilon^3 & 0 \end{pmatrix}.$$

Now let

$$D = \mathbb{F}_2[\pi], \qquad K = \mathbb{F}_2(\pi),$$

and define

$$A = D \otimes_k R, \qquad B = K \otimes_D A.$$

Note that the symbols π and ε have different meanings: π is the indeterminate in the domain D, while ε is nilpotent in S. In the polynomial ring $B[X]$, the indeterminate X commutes with all coefficients. Define

$$\Phi(X) = uX^2 + eX^3 + (e+u)X^4 + eX^5 + eX^6, \tag{2.2}$$

and set

$$P(X) = \frac{\Phi(X)}{\pi}, \qquad Q(X) = e. \tag{2.3}$$

We will prove

$$P, Q \in \mathrm{Int}(A) \qquad \text{but} \qquad PQ \notin \mathrm{Int}(A).$$

More precisely, the proof will establish

$$\Phi(r) = 0 \qquad \text{for every } r \in R,$$

while

$$(\Phi(X)e)(u+v) = s \neq 0.$$

The first identity implies that Φ/π is integer-valued on A. The second identity implies that $(\Phi/\pi)e$ is not integer-valued.

Theorem 2.1. *The domain $D = \mathbb{F}_2[\pi]$ has finite residue rings, and A is a torsion-free noncommutative D-algebra, free of rank 8, satisfying*

$$K \cap A = D$$

inside $B = K \otimes_D A$. Moreover,

$$P, Q \in \mathrm{Int}(A), \qquad PQ \notin \mathrm{Int}(A).$$

Consequently, $\mathrm{Int}(A)$ *is not a ring. Thus A gives the example requested in Problem 27a and disproves the conjecture in Problem 27b.*

Proof. We first verify that R is a finite noncommutative algebra.

Lemma 2.2. *The set R is a unital noncommutative $\mathbb{F}_2$-algebra of dimension 8. A $\mathbb{F}_2$-basis is*

$$e, f, u, v, p, q, s, w.$$

Consequently,

$$|R| = 2^8 = 256.$$

Proof. Write

$$S_{\mathrm{ev}} = \mathbb{F}_2 + \mathbb{F}_2\varepsilon^2, \qquad S_{\mathrm{odd}} = \mathbb{F}_2\varepsilon + \mathbb{F}_2\varepsilon^3.$$

Then R consists of matrices whose diagonal entries belong to S_{ev} and whose off-diagonal entries belong to S_{odd}. We have

$$S_{\mathrm{ev}}S_{\mathrm{ev}} \subseteq S_{\mathrm{ev}}, \qquad S_{\mathrm{odd}}S_{\mathrm{odd}} \subseteq S_{\mathrm{ev}},$$

and

$$S_{\mathrm{ev}}S_{\mathrm{odd}} \subseteq S_{\mathrm{odd}}, \qquad S_{\mathrm{odd}}S_{\mathrm{ev}} \subseteq S_{\mathrm{odd}}.$$

It follows from the usual formula for matrix multiplication that R is closed under multiplication. It is clearly closed under addition, and it contains

$$e + f = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus R is a unital subalgebra of $M_2(S)$. The eight elements

$$e, f, u, v, p, q, s, w$$

correspond to the eight independent coefficients in (2.1), so they form a $\mathbb{F}_2$-basis. Finally,

$$uv = p = \begin{pmatrix} \varepsilon^2 & 0 \\ 0 & 0 \end{pmatrix},$$

whereas

$$vu = q = \begin{pmatrix} 0 & 0 \\ 0 & \varepsilon^2 \end{pmatrix}.$$

Therefore $uv \neq vu$, so R is noncommutative. □

We next prove the crucial fact that Φ vanishes on every element of R.

Lemma 2.3. *For every $r \in R$,*

$$\Phi(r) = 0.$$

Proof. Take an arbitrary element

$$r = \begin{pmatrix} \alpha + \eta\varepsilon^2 & \gamma\varepsilon + \lambda\varepsilon^3 \\ \delta\varepsilon + \mu\varepsilon^3 & \beta + \theta\varepsilon^2 \end{pmatrix} \in R.$$

Set

$$\rho = 1 + \alpha + \beta, \qquad z = \eta + \theta.$$

We shall calculate

$$r^2 + r^4$$

and

$$r^3 + r^4 + r^5 + r^6.$$

Because S is commutative, the Cayley–Hamilton identity applies to the matrix r. Let

$$\tau = \mathrm{trace}(r) = \alpha + \beta + z\varepsilon^2$$

and

$$\Delta = \det(r) = \alpha\beta + (\alpha\theta + \beta\eta + \gamma\delta)\varepsilon^2.$$

Since the characteristic is 2, the Cayley–Hamilton identity becomes

$$r^2 = \tau r + \Delta I_2.$$

We also have

$$\tau^2 = \alpha + \beta, \qquad \Delta^2 = \alpha\beta,$$

because $\varepsilon^4 = 0$. Squaring the identity

$$r^2 = \tau r + \Delta I_2$$

and using $2 = 0$ in $\mathbb{F}_2$, we obtain

$$r^4 = \tau^2 r^2 + \Delta^2 I_2 = \tau^3 r + (\tau^2\Delta + \Delta^2) I_2.$$

It follows that

$$r^2 + r^4 = \tau(1 + \tau^2) r + \big(\Delta(1 + \tau^2) + \Delta^2\big) I_2.$$

Since

$$1 + \tau^2 = 1 + \alpha + \beta = \rho,$$

this calculation simplifies to

$$r^2 + r^4 = \rho \begin{pmatrix} \gamma\delta\varepsilon^2 & \gamma z\varepsilon^3 \\ \delta z\varepsilon^3 & \gamma\delta\varepsilon^2 \end{pmatrix}. \tag{2.4}$$

To see the simplication holds, if $\rho = 0$, then $\alpha \neq \beta$, and hence

$$\alpha\beta = 0.$$

The preceding Cayley–Hamilton expression is therefore zero. If $\rho = 1$, then $\alpha = \beta$. Write $\alpha = \beta = a$. Then $\tau = z\varepsilon^2$, and

$$\Delta = a + (az + \gamma\delta)\varepsilon^2.$$

Substitution into the preceding expression gives exactly (2.4).

Now observe that

$$r^3 + r^4 + r^5 + r^6 = (r + r^2)(r^2 + r^4).$$

If $\rho = 0$, then (2.4) shows that this expression is zero. Suppose that $\rho = 1$. Then $\alpha = \beta$. Every entry of $r^2 + r^4$ is divisible by ε^2. Therefore, when computing

$$(r + r^2)(r^2 + r^4),$$

only the terms of degree at most 1 in $r+r^2$ can contribute, because every term containing ε^4 is zero. Modulo $\varepsilon^2 M_2(S)$, we have

$$r + r^2 \equiv \begin{pmatrix} 0 & \gamma\varepsilon \\ \delta\varepsilon & 0 \end{pmatrix}.$$

Multiplying this matrix by the matrix in (2.4) gives

$$r^3 + r^4 + r^5 + r^6 = \rho \begin{pmatrix} 0 & \gamma\delta\varepsilon^3 \\ \gamma\delta\varepsilon^3 & 0 \end{pmatrix}. \tag{2.5}$$

We now return to the definition of Φ. Regrouping its terms gives

$$\begin{aligned} \Phi(r) &= ur^2 + er^3 + (e+u)r^4 + er^5 + er^6 \\ &= u(r^2 + r^4) + e(r^3 + r^4 + r^5 + r^6). \end{aligned}$$

Using (2.4), we obtain

$$u(r^2 + r^4) = \rho\gamma\delta \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix} = \rho\gamma\delta s.$$

Using (2.5), we obtain

$$e(r^3 + r^4 + r^5 + r^6) = \rho\gamma\delta \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix} = \rho\gamma\delta s.$$

Therefore

$$\Phi(r) = \rho\gamma\delta s + \rho\gamma\delta s.$$

Since the characteristic is 2,

$$x + x = 0$$

for every $x \in R$. Hence

$$\Phi(r) = 0.$$

Since r was arbitrary, Φ vanishes on all of R. $\square$

Now we show that, although Φ vanishes everywhere, multiplying it on the right by the constant polynomial e destroys this property.

Lemma 2.4. $(\Phi(X)e)(u+v) = s \neq 0$.

Proof. Since X commutes with the coefficients,

$$\begin{aligned} \Phi(X)e = (ue)X^2 + (e^2)X^3 + ((e+u)e)X^4 \\ + (e^2)X^5 + (e^2)X^6. \end{aligned}$$

The relevant multiplication rules are

$$ue = 0, \qquad e^2 = e, \qquad (e+u)e = e.$$

Therefore

$$\Phi(X)e = eX^3 + eX^4 + eX^5 + eX^6. \tag{2.6}$$

Now let

$$a_0 = u + v.$$

Because $u^2 = v^2 = 0$, we have

$$a_0^2 = (u+v)^2 = uv + vu = p + q.$$

Consequently,

$$\begin{aligned} a_0^3 &= (p+q)(u+v) \\ &= pu + pv + qu + qv \\ &= s + w. \end{aligned}$$

Indeed,

$$pu = s, \qquad qv = w, \qquad pv = qu = 0.$$

Also we know $a_0^4 = 0$ because every term in a_0^4 contains ε^4. Thus

$$a_0^5 = a_0^6 = 0.$$

Using (2.6), we get

$$\begin{aligned} (\Phi(X)e)(a_0) &= ea_0^3 + ea_0^4 + ea_0^5 + ea_0^6 \\ &= e(s+w). \end{aligned}$$

Since

$$es = s, \qquad ew = 0,$$

we conclude that

$$(\Phi(X)e)(a_0) = s.$$

Finally,

$$s = \begin{pmatrix} 0 & \varepsilon^3 \\ 0 & 0 \end{pmatrix} \neq 0,$$

because $\varepsilon^3 \neq 0$ in

$$S = \mathbb{F}_2[\varepsilon]/(\varepsilon^4).$$

□

We are now ready to prove the full Theorem.

Proof of Theorem 2.1. By Lemma 2.2,

$$A = De \oplus Df \oplus Du \oplus Dv \oplus Dp \oplus Dq \oplus Ds \oplus Dw.$$

Therefore A is a free D-module of rank 8. In particular, A is finitely generated and torsion-free over D. The scalar embedding is

$$D \longrightarrow A, \qquad d \longmapsto d(e+f).$$

Since $e+f$ is the identity of R, the image of D lies in the center of A. We also have

$$B = K \otimes_D A \cong K \otimes_k R.$$

Suppose that a scalar $\kappa \in K$ belongs to A. As an element of B, this scalar is

$$\kappa(e+f) = \kappa e + \kappa f.$$

In the displayed D-basis of A, the coefficients of both e and f are κ. Since coefficients of elements of A must lie in D, we obtain $\kappa \in D$. Thus we have $K \cap A = D$.

Next, $D = \mathbb{F}_2[\pi]$ has finite residue rings. Indeed, let

$$0 \neq d(\pi) \in D$$

have degree m. Every residue class in D/dD has a representative of degree less than m. There are exactly 2^m such polynomials. Therefore D/dD is finite. If d is a unit, the quotient is the one-element zero ring.

It remains to prove the statements concerning P and Q. Reduction modulo π gives

$$A/\pi A \cong (D/\pi D) \otimes_k R \cong k \otimes_k R \cong R.$$

For $a \in A$, denote its image in R by $\overline{a}$. Reduction modulo π commutes with addition, multiplication, powers, and polynomial evaluation. Therefore

$$\overline{\Phi(a)} = \Phi(\overline{a}).$$

By Lemma 2.3,

$$\Phi(\overline{a}) = 0.$$

Hence

$$\overline{\Phi(a)} = 0.$$

The kernel of the reduction map

$$A \longrightarrow A/\pi A$$

is πA. Thus

$$\Phi(a) \in \pi A.$$

Consequently,

$$P(a) = \frac{\Phi(a)}{\pi} \in A.$$

This holds for every $a \in A$, so $P \in \mathrm{Int}(A)$. The polynomial $Q = e$ is constant and has value in A. Therefore $Q \in \mathrm{Int}(A)$. Finally, evaluate the product at

$$a_0 = u + v.$$

Since

$$P(X)Q(X) = \frac{\Phi(X)e}{\pi},$$

Lemma 2.4 gives

$$(PQ)(a_0) = \frac{(\Phi(X)e)(a_0)}{\pi} = \frac{s}{\pi}.$$

We claim that

$$\frac{s}{\pi} \notin A.$$

Every element of A has a unique expression

$$d_1 e + d_2 f + d_3 u + d_4 v + d_5 p + d_6 q + d_7 s + d_8 w,$$

with $d_1, \ldots, d_8 \in D$. The coefficient of s in s/π is $1/\pi$. But

$$\frac{1}{\pi} \notin \mathbb{F}_2[\pi].$$

Indeed, if

$$\frac{1}{\pi} = g(\pi)$$

for some $g(\pi) \in \mathbb{F}_2[\pi]$, then $\pi g(\pi) = 1$. The left-hand side has constant term 0, while the right-hand side has constant term 1, which is impossible. Therefore

$$\frac{s}{\pi} \notin A.$$

It follows that $(PQ)(a_0) \notin A$, and hence $PQ \notin \mathrm{Int}(A)$. Thus $\mathrm{Int}(A)$ is not closed under multiplication. $\square$

$\square$

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